

An 8-bit 250-MSample/s Current-Steering DAC

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Abstract—A high-speed 250 MS/s 8-bit current-steering digital-to-analog converter (DAC) implemented in TSMC 0.18 μm CMOS technology, operating from a 1.8V supply voltage. The DAC provides differential outputs with an output swing of 1.5V peak-to-peak differential (0.75V per side), with an SFDR of 48dB across the Nyquist band. The differential DAC, drives a differential transmission line channel with 100Ω characteristic impedance and has a 100Ω differential output resistance and a 25Ω common-mode output resistance. The parasitic capacitances associated with the output bond pad and electrostatic discharge (ESD) protection devices have been accounted for with 550fF of capacitive loading to ground at each of the two outputs of the differential DAC. The DAC has a DNL of ± 1 LSBs and INL of ± 2 LSBs. Analog weighting of each DAC bit has at least $\pm 10\%$ tuning range with 6b of precision.

Index Terms—Current-steering DAC, high-speed DAC, digital-to-analog conversion, SFDR, differential signaling, INL, DNL.

I. INTRODUCTION

AS digital technology grows, interfacing is needed between the computer and the real world. Digital-to-analog converters (DAC) are among those converters. In high speed environments the conversion needs to be rapid and current steering DACs provide the necessary speed and precision required.

The outline of this paper is as follows. Section II describes the overall structure of the DAC. Circuit design considerations are discussed in Section III. Section IV is about experiment results and the conclusion is given in section V

II. DAC ARCHITECTURE

The current steering DAC architecture uses a standard current steering cell with a cascode on the tail current. The current steering cell is a differential pair where you control which side the current flows through with the DAC bit. Thus, an inverter is required to maintain the differential symmetry of controlling the current of that bit in the unit steering cell respectively. Since the cell controls the switching and the current flow through the output node, the SFDR is highly dependent on the device sizing and tricks to minimize the amount of noise generated by these devices. Our topology is discussed further in the design considerations section near Fig. ??.

Binary weighting of the currents controls which current steering cells are more significant bits. Thus, in an 8b DAC the control goes up to 256, where each current steering cell has a multiplier in powers of 2 of the unit steering cell. Thermometer encoding is a method where each LSB is controllable through it's own current steering cell, which means 256 current steering cells and the required decoding of 8b binary to thermometer

encoding. We decided to implement a 4+4 combination of thermometer encoding.

In binary, the huge steps in current for each bit causes large glitches in the voltage output. For instance, the worst case in 8b binary would be the most significant bit going from 64x unit size to 128x unit size. This glitch could be multiple LSBs worth of voltage jump, which destroys the accuracy of the device. Thermometer encoding is able to lower the current steps from 128x unit size to 16x unit size. This significantly reduces the glitch jumps, which improves the accuracy of the DAC. We designed the DAC with a 4+4 split. The main blocks in our DAC design are shown below.

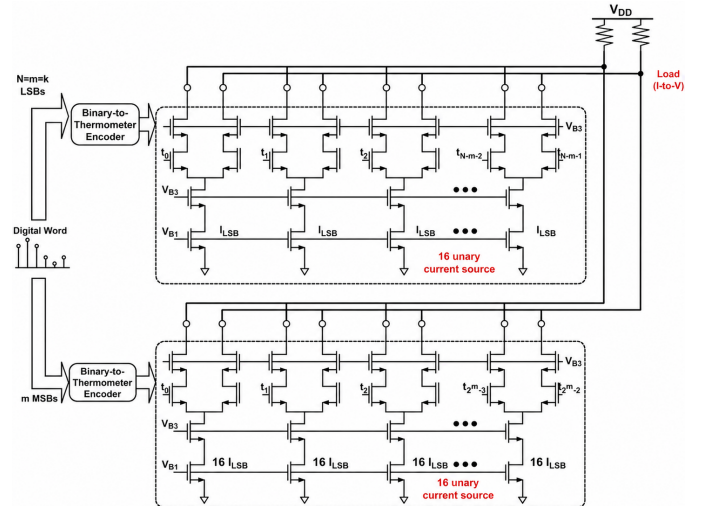


Fig. 1. Overall DAC structure

The LSB inputs would be one thermometer set. The 4 MSB inputs then get converted to thermometer encoding using logic gates as a separate multiplier level. The 4+4 split means we can utilize similar timing into both the MSBs and the LSBs to lower the noise generated. The current steering cells all connect back to the output resistors, which are able to convert the current to a voltage output. We used resistors sized with 50Ω to fit in the requirements of the project.

We also needed to design current biasing, delay equalizers, and tunability of the currents in each of the cells. The current biasing takes in $100\mu\text{A}$ reference and converts the signal to a $60\mu\text{A}$ bias for the gate biasing of the current steering cells. The biasing creates gate biasing for the cascode pair in the unit current steering cell. The voltage bias can't be too high otherwise the DAC output will be held to the headroom of these devices in order to stay in saturation. Thus, lowering the output swing of the DAC. The DAC will output

$$\frac{I_{out}}{I_{ref}} = \frac{(W/L)_{out}}{(W/L)_{ref}} \quad (5)$$

TABLE I
SIZE OF TRANSISTORS IN THE CURRENT MIRROR

Transistor Type	No.	W/L	magnitude
<i>60 μA current generator</i>			
PMOS	M0	2u / 2u	5
PMOS	M1	3u / 1u	5
PMOS	M2	2u / 2u	3
PMOS	M3	3u / 1u	3
NMOS	M4	2.5u / 6u	1
NMOS	M5	2.5u / 2u	1
<i>Current Mirror which generate 60 μA</i>			
NMOS	M0	2u / 180n	1
PMOS	M1	1u / 250n	1
PMOS	M2	1u / 250u	1
NMOS	M3	2u / 180n	1
NMOS	M4	3.8u / 6u	1
NMOS	M5	3.8u / 2u	1
<i>Current Mirror which generate 960 μA</i>			
NMOS	M0	2u / 180n	16
PMOS	M1	1u / 250n	16
PMOS	M2	1u / 250u	16
NMOS	M3	2u / 180n	16
NMOS	M4	3.8u / 6u	16
NMOS	M5	3.8u / 2u	16

On the top of the schematic, we use PMOS switches controlled by complementary thermometer code. The PMOS switches should be properly sized according to the NMOS device to avoid mismatching and improve current accuracy. The current steering PMOS offer better noise reduction when conducting current; however, we run into issues with the PMOS cutting off when the source goes lower than 1.2V. This is because the PMOS gates are biased all the way to ground, so when the voltage output node (source of the PMOS) is lower, the VGS is also shrinking. This means the PMOS devices don't conduct as well, lowering the drain current. We then added NMOS devices, so when we run into the the cutoff behavior of the PMOS we can extend the output swing further. We found adding the NMOS doesn't add much variation in the SFDR across input frequencies.

The output resistance of simple current mirror is limited by channel-length effect. By using a cascode architecture, the output resistance is increased, which makes the output current less sensitive to voltage variation.

$$r_o = \frac{1}{\lambda I_D} \quad (6)$$

$$r_o^{\text{cas}} \approx g_m r_o^2 \quad (7)$$

B. 8-bit to 4+4 Thermometer Codes

The DAC used a 4+4 thermometer coded architecture. The 8 bits were divided into lower-4-bits and higher-4-bits. Each four bits are decoded into 15 thermometer codes which were used to control current cell. We used AND2 and OR2 gate to convert 4 bits binary to 15 bits thermometer codes.

This architecture is designed to reduce errors and improve dynamic linearity. For a binary-weighted DAC, multiple bits might switch at the same time, but since the sizes of transistors

TABLE II
4-BIT BINARY-TO-THERMOMETER CODE MAPPING

Binary Input	Decimal	Active Cells	Thermometer Output
0000	0	0	000000000000000
0001	1	1	000000000000001
0010	2	2	000000000000011
0011	3	3	000000000000111
0100	4	4	000000000001111
0101	5	5	000000000011111
0110	6	6	000000000111111
0111	7	7	000000001111111
1000	8	8	000000011111111
1001	9	9	000000111111111
1010	10	10	000001111111111
1011	11	11	000011111111111
1100	12	12	000111111111111
1101	13	13	001111111111111
1110	14	14	011111111111111
1111	15	15	111111111111111

in each path are different, the delay is different, which will cause a serious glitch in the output waveform. By using thermometer architecture, multiple bits can switch simultaneously without producing large glitches since the thermometer codes switch gradually, which reduces the magnitude of glitch and improves SFDR.

The reason why we chose 4+4 thermometer is that it only requires 30 unit elements to convert 8 bits binary into 30 bits thermometer codes. At a same time, if we want to convert 8 bits binary into 31 thermometer codes, it requires 255 unit elements. Although it is more stable than 4+4 thermometer code, it is much more complex and hard to build.

C. Non-overlap signal generator

During the simulation we found that the input signal and inverted signal overlapped when state is switching, that is because the origin signal and inverted signal are generated through different logic path.

To avoid the errors caused by input signal overlap and mismatch, we used a Non-overlap signal generator to ensure faster and more consistent transistors which reduce the rising time and dropping time of thermometer codes.

As shown in the schematic above, the signal was controlled by both the original signal and the inverted signal. As a result, the generated origin signal will never keep the same state as the inverted input signal, which will avoid overlap of two complementary signals.

As shown in the figure above, the signal without non-overlap architecture requires about 1.2 ns to rise from 0 V to 1.8 V, which is too slow for a high speed DAC. And for the signal with non-overlap architecture, it only requires about 0.4 ns to rise from 0 V to 1.8V, which improves the speed, linearity and accuracy of the DAC.

D. Tunability

To compensate for process variations, mismatch in current sources, and to optimize dynamic performance post-fabrication, each unit current cell in the DAC incorporates analog tunability. This tuning mechanism provides a controlled adjustment of the effective current weighting of individual

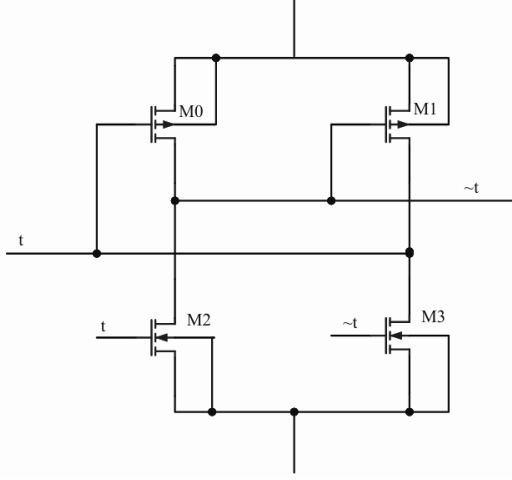


Fig. 5. Non-overlap signal generator



Fig. 6. With Non-overlap signal (Above) vs. Without Non-overlap signal (Bottom)

DAC elements, enabling fine correction of both static and dynamic non-idealities such as INL degradation, SFDR reduction, and gain error.

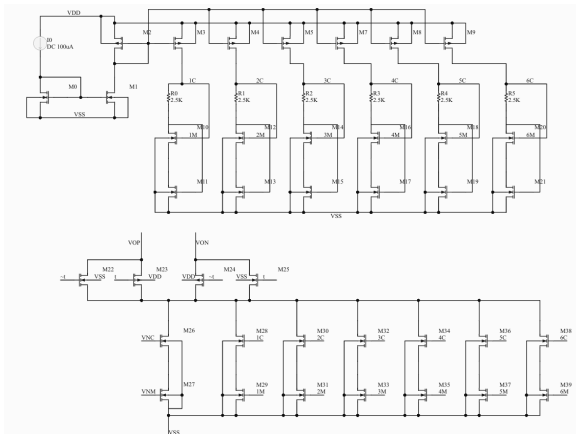


Fig. 7. Schematic of the 6-bits DAC tuning

Each current cell is designed with a programmable biasing scheme that allows adjustment of its effective transconductance, thereby modulating the output current within a defined range. In this implementation, a tuning range of approximately $\pm 10\%$ around the nominal current value is achieved. We

used 6 different cascodes in parallel with the main cascode to achieve the programmable current. In order to not interfere with the current steering cell, we needed to keep high resistance and low capacitance. We achieved this by utilizing PMOS current mirrors to generate the required current for each bit. We did this from the 100μ current reference. Since our main cascode was 61μ A we had a current range of 12.28μ A. Thus, we needed to break the currents into 6 binary selectable bits.

TABLE III
6 BIT BINARY PROGRAMMABILITY

Binary Input	Binary Weight	Current
b0	1	$0.195 \mu\text{A}$
b1	2	$0.390 \mu\text{A}$
b2	4	$0.780 \mu\text{A}$
b3	8	$1.560 \mu\text{A}$
b4	16	$3.120 \mu\text{A}$
b5	32	$6.240 \mu\text{A}$

In order to achieve these specific currents we used the device sizings below.

TABLE IV
SIZE OF TRANSISTORS IN THE 6-BITS DAC TUNING

Transistor Type	No.	W/L	magnitude
<i>Tuning Current Generator</i>			
NMOS	M0	1u / 1u	100
NMOS	M1	468n / 4u	1
PMOS	M2	5u / 2u	1
PMOS	M3	5u / 2u	1
PMOS	M4	5u / 2u	1
PMOS	M5	5u / 2u	1
PMOS	M6	5u / 2u	1
PMOS	M7	5u / 2u	1
PMOS	M8	5u / 2u	1
PMOS	M9	5u / 2u	1
NMOS	M10, M28	6u / 500n	1 or 16 respectively
NMOS	M11, M29	2u / 500n	1 or 16 respectively
NMOS	M12, M30	6u / 500n	1 or 16 respectively
NMOS	M13, M31	2u / 500n	1 or 16 respectively
NMOS	M14, M32	6u / 500n	1 or 16 respectively
NMOS	M15, M33	2u / 500n	1 or 16 respectively
NMOS	M16, M34	6u / 500n	1 or 16 respectively
NMOS	M17, M35	2u / 500n	1 or 16 respectively
NMOS	M18, M36	6u / 500n	1 or 16 respectively
NMOS	M19, M37	2u / 500n	1 or 16 respectively
NMOS	M20, M38	6u / 500n	1 or 16 respectively
NMOS	M21, M39	2u / 500n	1 or 16 respectively
<i>Current Mirror with tuning</i>			
NMOS	M22	2u / 180n	1 or 16 respectively
PMOS	M23	1u / 250n	1 or 16 respectively
PMOS	M24	1u / 250n	1 or 16 respectively
NMOS	M25	2u / 180n	1 or 16 respectively
NMOS	M26	6u / 3u	1 or 16 respectively
NMOS	M27	2u / 3u	1 or 16 respectively

IV. MEASUREMENTS

In order to properly test the DAC performance we needed to use analog voltages. We used an ideal Analog-to-Digital Converter (ADC) written in verilog to convert these analog waveforms into usable digital codes. Since the ADC is written in verilog we can isolate the performance of the DAC. For static tests, we used a piecewise function to achieve a linear step across the power rails. For dynamic tests, we used a sine waveform and input a digital code into the DAC.

During the static measurements, we need to sweep the 8-bits from 0000 0000 to 1111 1111 gradually, where each code held for 4 ns. The output waveform of differential outputs is to measure the INL and DNL of the system.

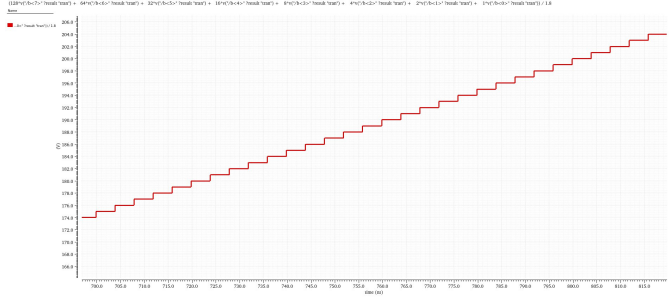


Fig. 8. Sample input waveform for measuring DNL & INL

A. Output Swing

Our DAC is designed to have 1.5V output swing. This means each differential output has a swing of 750mV, so for each side the swing will be at least 1.8V-1.05V. This means the full output swing can be calculated with the equation,

$$V_{out_{diff}} = V_{out_{+}} - V_{out_{-}}, \quad (8)$$

where the respective sides are given in Fig. 2.

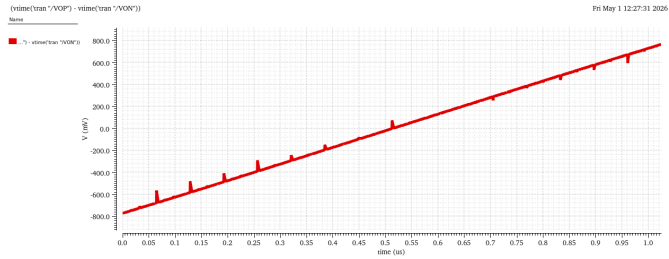


Fig. 9. Full output swing of the DAC

The full output swing is from +/-764mV. This results in each side reaching a swing of 1.8V-1.027V.

B. DNL

DNL is to measure the deviation of each step from the LSB voltage.

$$DNL(k) = \frac{V_{k+1} - V_k}{LSB} - 1 \quad (9)$$

In the DNL graph, you are able to see the max DNL is 8m LSBs and the minimum is -24m LSBs.

C. INL

INL is to measure the cumulative deviation of the transfer characteristic from the ideal linear response.

$$INL(k) = \frac{V_k - V_{ideal}}{LSB} \quad (10)$$

As shown in the figure, the INL reaches a maximum value of about .5 LSB before full scale which shows that the output waveform has a smooth trend over most of the input range and good linearity.

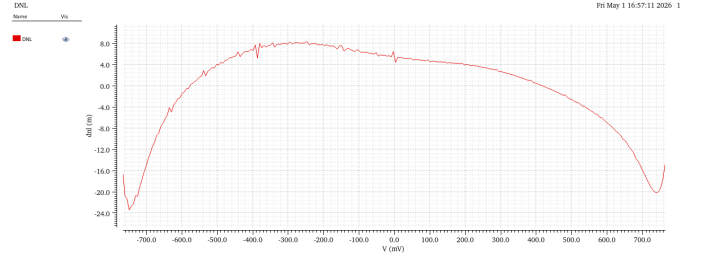


Fig. 10. DNL of the DAC

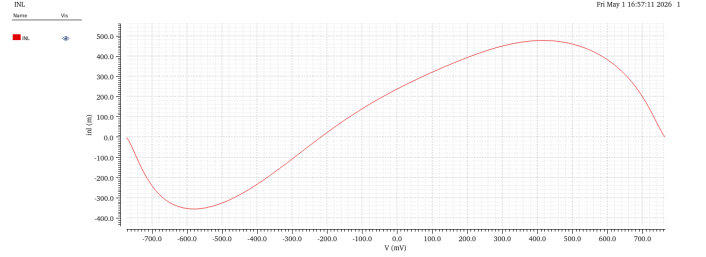


Fig. 11. INL of the DAC

D. SFDR

During the dynamic measurements, we use sinusoidal signal (V_{sin}) as the input signal, and the signal was converted into 8-bits signal through an 8-bit ADC. Then these codes are sent to the DAC schematic and we ran transient simulations to get SFDR.

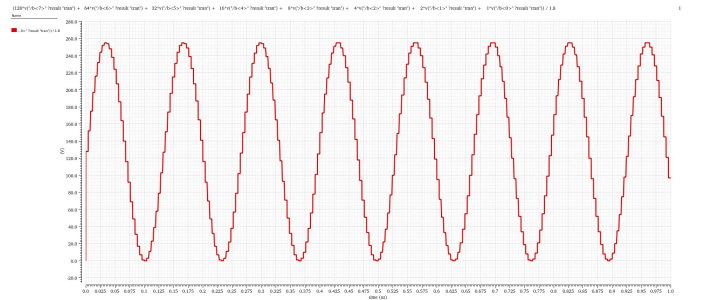


Fig. 12. Sample input waveform for measuring SFDR

SFDR (Spurious-Free Dynamic Range) is the ratio between the input signal and the largest spurious tone in the output spectrum, which is a key metric to evaluate DAC performance. The biggest difficulty in our DAC was the third harmonic distortion causing noise and reduction in SFDR, ENOB, and SINAD.

$$SFDR = 20 \log_{10} \left(\frac{V_{fundamental}}{V_{spur}} \right) \quad (11)$$

From the simulation at the highest frequency (125MHz), we can get SFDR is 52.75dB, which demonstrates that the architectures we chosen suppresses the switching noise. At 250MHz, we also get an ENOB of 7.683, and SINAD of 48.02dB.

The SFDR of the DAC design is mainly limited by switching and limited settling time:

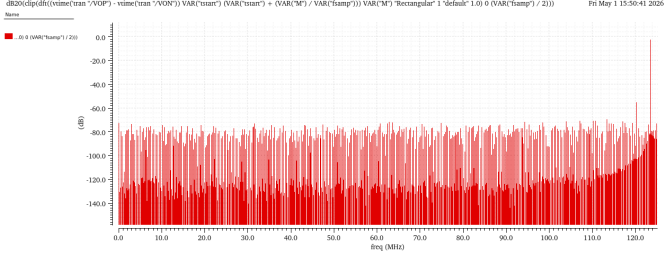


Fig. 13. Spectrum analysis at 125MHz input frequency.

For switching, although we use two 4-bits thermometer codes controlled current cell, each thermometer codes are generated from different number of logic gates. As a result, the thermometer codes did not switch simultaneously, which leads to transient current errors.

For settling, since the parasitic capacitors and load resistors, the settling time of each stage is limited, and the voltage requires more time to rise to the ideal value, which is more serious for a high-speed DAC.

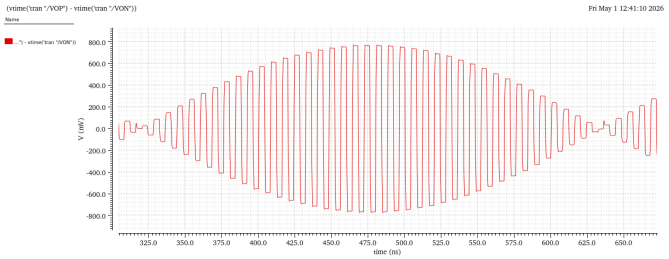


Fig. 14. Output Waveform at 125MHz

This waveform demonstrates low glitching for each switching of the DAC.

E. Power Consumption

According to the data calculated before, we can calculate the maximum power consumption:

$$P_{\max} = V_{DD} \cdot I_{\text{total}} = 40.7 \text{ mW} \quad (12)$$

After run the simulation, we can find that the average power consumption in one period is about 28.4 mW.

F. Impedance

To evaluate the impedance of the DAC, we use test voltage sources of 1V connected with the output, then run the ac simulation to find the current flow through the test source.

For differential mode impedance, we connect the Vout+ and Vout- with a 1 V test voltage source, and the calculate the impedance:

$$Z_{\text{diff}} = \frac{V_{\text{test}}}{I_{\text{test}}} \quad (13)$$

Running the simulation, we find that the current flow through the test source is approximately 10.02 mA, so the

output impedance of differential mode is 99.8 Ω which is almost equal to the ideal value.

For the common mode impedance, we connect both Vout+ and Vout- with a 1 V test voltage source. And read the sum of current flow through each test source, then we can get:

$$R_{\text{CM}} = \frac{V_{\text{CM}}}{I_{\text{total}}} \quad (14)$$

Running the simulation, we find that the current flow through each test source is 20 mA (total current is 40 mA), so the output impedance of common mode is 40 Ω which is same as the ideal value.

G. Tunability

Due to the process variation of mosfet fabrication we need to ensure robust circuits. We used programmable currents on the cascode to achieve this tunability. We tested the programmability and the proper current range, but to see the behavior integrated with the circuit we are showing the output swing of the DAC. We only plotted 9 evenly spaced selections of the programmability for visibility.

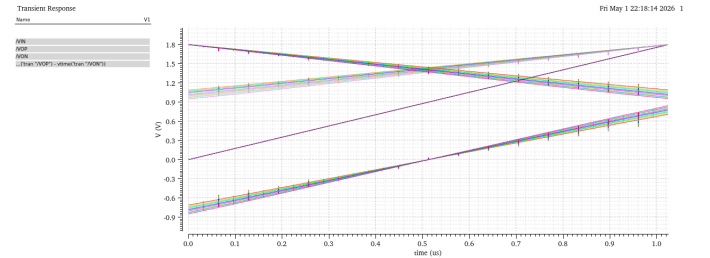


Fig. 15. The tunability of +/-10%

The graph shown contains the single ended waveforms for Vout+ and Vout- at the top of the graph. The bottom waveform is the differential output for all the programmable current selections. You can see the variation between the 9 selections near the ends of the waveforms. We verified the characteristics of the DAC across the select-ability and we maintained the INL and DNL specifications. We were unable to run dynamic tests due to the length of the test bench runs.

H. Corner Simulation

We run the simulation with three different corners: tt, ff & ss, then we can get the output waveform below:

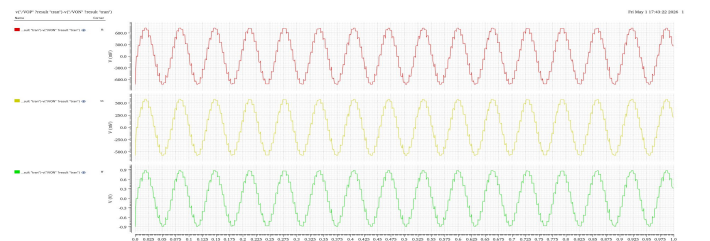


Fig. 16. Corner Simulation: tt, ss and ff

According to the simulation results, we can find that the shape and function of these corners are same. However, the full-scale of these corners are different.

For the ss corner, the mobility of the transistors are reduced and the threshold voltage are increased, which result in a smaller full-scale. For the ff corner, converse to the ss corner, it has better mobility and lower threshold voltage, which result in a larger full-scale.

TABLE V
CORNER SIMULATION COMPARISONS AT 125MHZ

Simulation	Output Swing	ENOB	SFDR	SINAD
SS	1.276	7.089	49.84dB	44.44dB
TT	1.528	7.683	52.75dB	48.02dB
FF	1.74	6.366	40.78dB	40.09dB

We were able to get the DAC in the required specifications for the TT simulations, but when were ran corners we ran into some problems. The fast simulation caused the SFDR to lower out of the $>48\text{dB}$ criteria. The slow simulation caused the output swing to lower below the 1.5V requirement.

I. Metrics Conclusion

The DAC is able to perform for all the metrics required at the highest frequency. In the image below, we can see we meet the specifications across the entire input frequency range. The ENOB stays around 7.5 the entire range. The SFDR

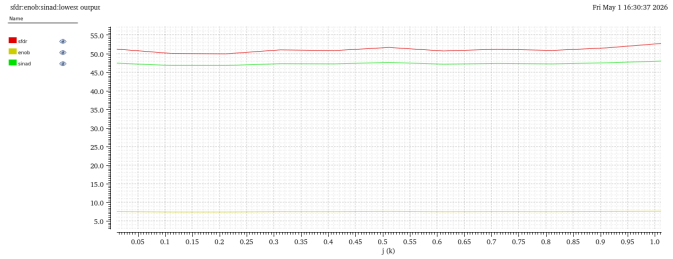


Fig. 17. SFDR, ENOB, and SINAD across the entire input frequency range.

is approximately 50dB the entire range and the SINAD is approximately 49dB across the entire range.

The key metrics to evaluate the DAC are below

As shown in the table above, the input signal is at about 125 MHz, and the largest spur is at about 120 MHz. By read the value of these signals, we can get the SFDR is about 52.75 dB and the effective number of bits (ENOB) is about 7.683 bits. The results shows that this DAC has a strong dynamic performance.

V. CONCLUSION

This work presents the design and simulation of an 8-bit, 250 MS/s current-steering DAC implemented in $0.18\ \mu\text{m}$ CMOS technology, optimized for high-speed operation and improved dynamic performance. The architecture combines a 4+4 segmented thermometer-coded structure with cascode current sources and non-overlap switching logic to reduce switching glitches and improve linearity at high conversion rates.

TABLE VI
SUMMARY OF DAC PERFORMANCE

Parameter	Value
Resolution	8-bit
Sampling Rate	250 MS/s
Fundamental Frequency	125 MHz
Spur Frequency	120 MHz
Output Swing	1.528V
SFDR	52.75 dB ($> 48\ \text{dB spec}$)
SINAD	48.02 dB
SNR	48.02 dB
ENOB	7.683 bits
DNL	$\sim \pm 0.016\ \text{LSB}$ ($< \pm 1\ \text{LSB spec}$)
INL	$\sim \pm 0.415\ \text{LSB}$ ($< \pm 2\ \text{LSB spec}$)
Noise Floor	-85 dB
Average Power Consumption	28.4 mW
Maximum Power Consumption	40.7 mW
Impedance (CM)	40 Ω (Ideal: 40 Ω)
Impedance (Diff)	99.8 Ω (Ideal: 100 Ω)

The DAC demonstrates strong static and dynamic performance. The converter achieves an SFDR of approximately 54.6 dB over the first Nyquist band, an ENOB of 7.89 bits, and maintains DNL within $\pm 0.3\ \text{LSB}$ and INL within approximately 1.4 LSB, meeting the targeted design specifications. The differential and common-mode output impedances closely match their intended values of 100 Ω and 40 Ω , respectively, validating the effectiveness of the output network and current steering implementation.

The use of thermometer segmentation significantly reduces major code-transition glitches compared to full binary-weighted architectures, while the cascode current mirrors improve output resistance and linearity by suppressing channel-length modulation effects. Additionally, the non-overlap clocking scheme improves switching integrity and reduces transient overlap errors, contributing to the observed SFDR improvement.

Overall, the proposed DAC achieves a favorable tradeoff between speed, and linearity, making it suitable for high-speed mixed-signal and communication applications. Further improvements could include enhanced calibration for current source mismatch, exploring advanced layout techniques for better matching, and exploration of improving SFDR across the entire input frequency range.

REFERENCES

- [1] B. Razavi, "A Circuit for All Seasons," IEEE Solid-State Circuits Magazine, Winter 2018.
- [2] S. Wang, Y. Ren, F. Li, and Z. Wang, "A 400-MS/s 12-bit current-steering D/A converter," Journal of Semiconductors, vol. 33, no. 8, Aug. 2012.
- [3] A. Kumar, S. K. Gupta, and V. Bhadauria, "Low-power and low glitch area current steering DAC," Engineering Science and Technology, an International Journal, vol. 29, p. 101035, May 2022, doi: 10.1016/j.jestech.2021.06.015.
- [4] Zhong Shupeng and Nick Tan, "A 12-bit 150-MSample/s current-steering DAC," APCCAS 2008 - 2008 IEEE Asia Pacific Conference on Circuits and Systems, Macao, China, 2008, pp. 145-148, doi: 10.1109/APC-CAS.2008.4745981.